

Esercizio 11.4

giovedì 7 ottobre 2021 10:18

$$A = a \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \end{pmatrix}$$

X
Y

$$|4\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$A^\dagger = A$$

$$|A - \alpha I| = 0$$

$$-\alpha x_1 - i x_2 = 0 \quad \alpha = \pm 1$$

$$-x_1 - i x_2 = 0 \quad x_1 = -i x_2$$

$$|4\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \quad x_2 = i x_1$$

$$x_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \end{pmatrix} = |x_- \rangle$$

$$|y_+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 0 \\ 1 \\ i \end{pmatrix} \quad |y_-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 0 \\ 1 \\ -i \end{pmatrix}$$

$|x_+\rangle, |y_+\rangle$ degenerati
 $|x_-\rangle, |y_-\rangle$

$$\langle x_+ | y_+ \rangle = 0$$

$$P^{+a} = |\langle x_+ | \psi \rangle|^2 + |\langle y_+ | \psi \rangle|^2 =$$

$$= \left| \frac{1+1}{2\sqrt{2}} \right|^2 + \left| \frac{1-i}{2\sqrt{2}} \right|^2 =$$

$$= \frac{1}{8} (4 + 2 + 2) = \frac{6}{8} = \frac{3}{4}$$

$$P^{-a} = |\langle x_- | \psi \rangle|^2 + |\langle y_- | \psi \rangle|^2 = \cancel{\left| \frac{1-1}{2\sqrt{2}} \right|^2} + \left| \frac{1+i}{2\sqrt{2}} \right|^2 = \frac{1}{4}$$